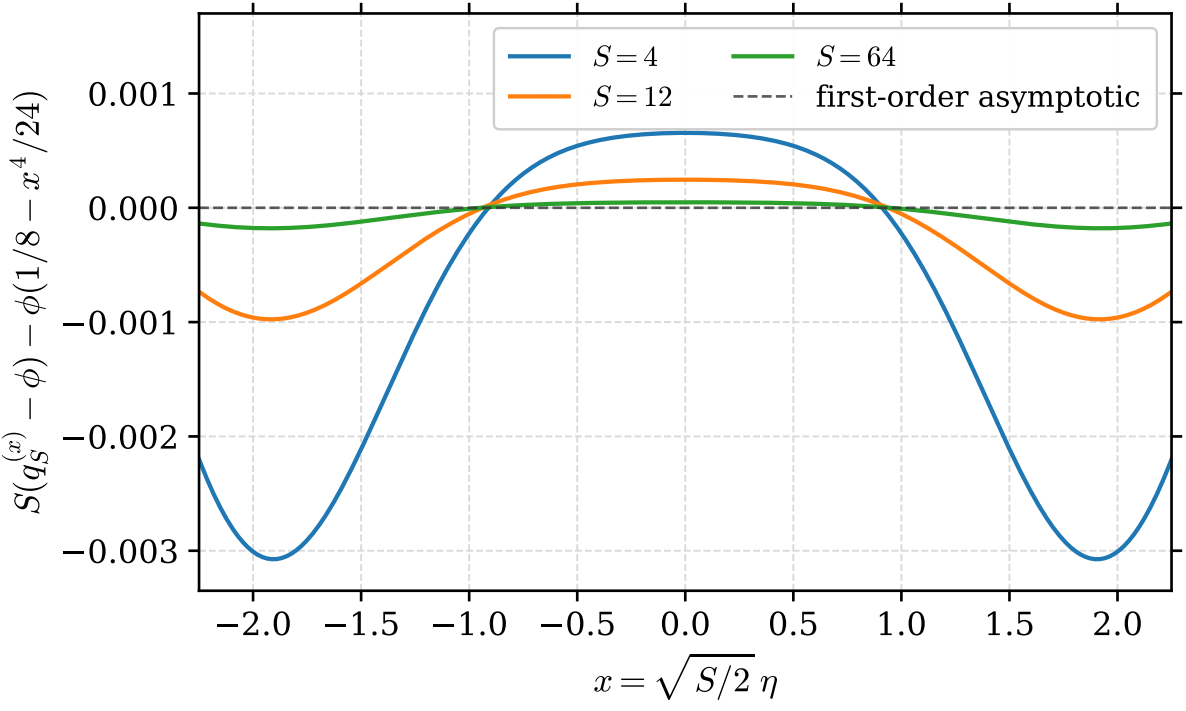
$S=4$
$S=12$
$S=64$
first-order asymptotic
0.001
0.000
−0.001
−0.002
−0.003
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
$S(q_S^{(x)} - \phi) - \phi(1/8 - x^4/24)$
$x = \sqrt{S/2}\,\eta$